

that enable us to say that phi is equal to x/y . Moreover, no one has yet been able to discern any pattern in its digits.

We have an infinite decimal expansion with no clear pattern and no fractional representation. It may seem impossible to ever clearly express phi's exact value. But if we learn more about phi, we can find a way to express it both exactly and concisely. One of the things we know about phi is that it's the solution to this equation:

$$\phi^2 - \phi - 1 = 0$$

One way we might imagine expressing the exact value of phi would be to write "the solution to the equation written above this paragraph." This has the benefit of being concise and technically exact, but it means that we have to solve the equation somehow. That description also doesn't tell us the 200th or 500th digit in phi's expansion.

If we divide our equation by phi, we get the following:

$$\phi - 1 - \frac{1}{\phi} = 0$$

And if we rearrange that equation, we get this:

$$\phi = 1 + \frac{1}{\phi}$$

Now imagine if we attempted a strange substitution of this equation into itself:

$$\phi = 1 + \frac{1}{\phi} = 1 + \frac{1}{1 + \frac{1}{\phi}}$$

Here, we rewrote the phi on the righthand side as $1 + 1/\phi$. We could do that same substitution again; why not?